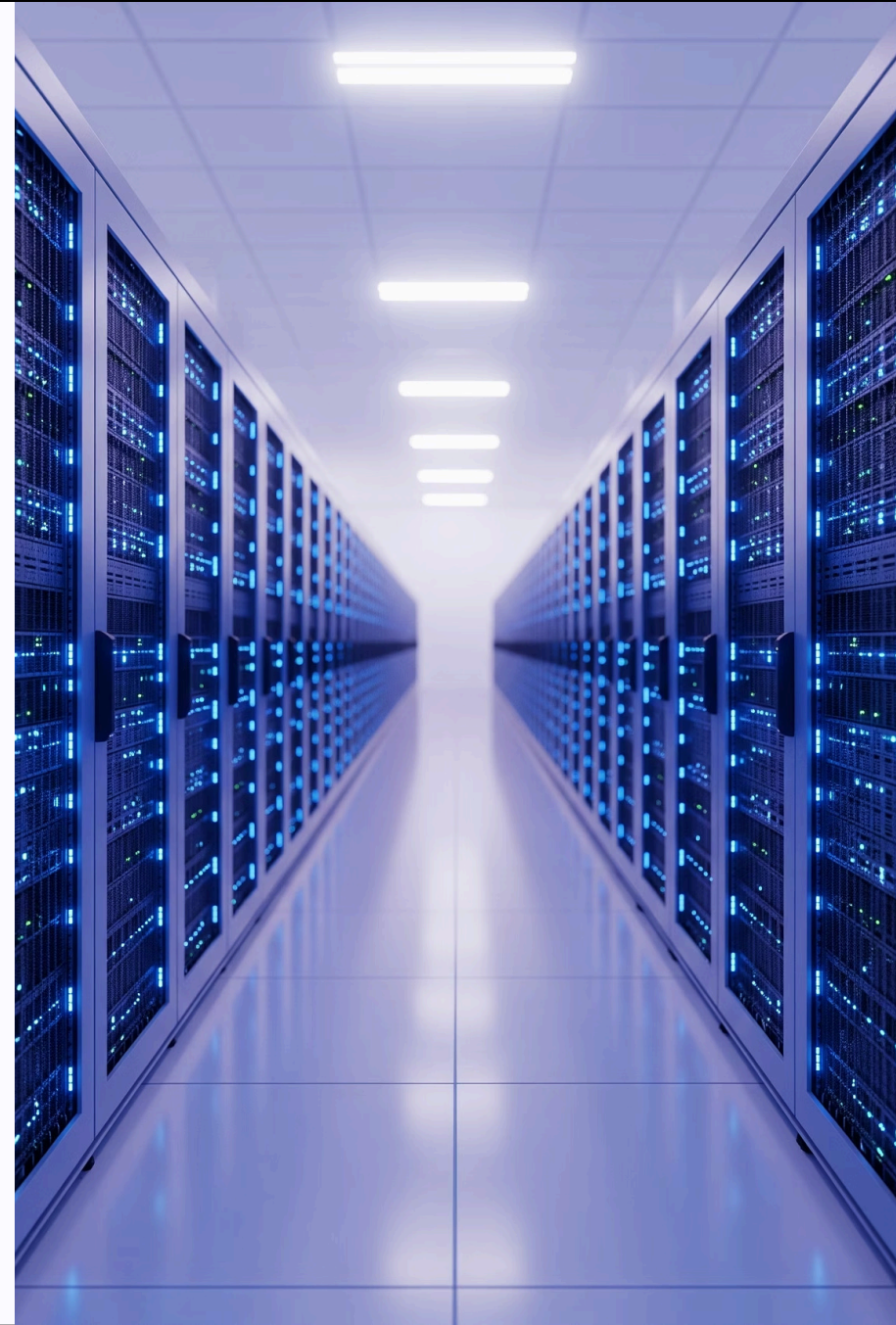


Storage Cost Optimization Strategies

Balancing Performance, Scale, and Budget

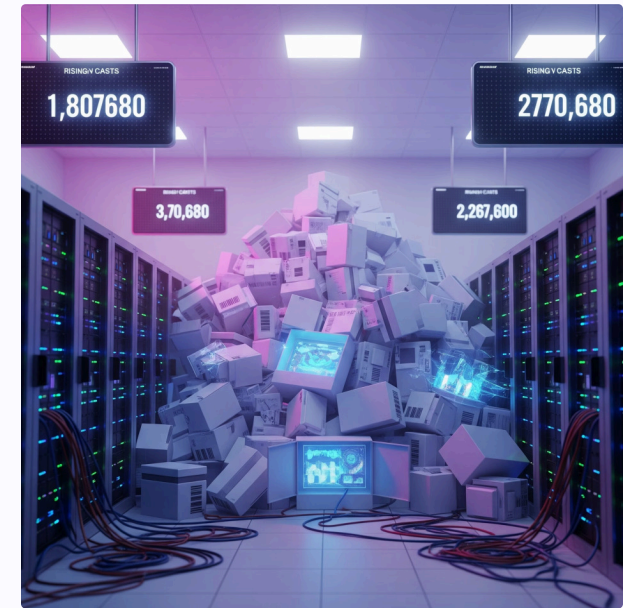
A comprehensive approach to managing your AI infrastructure costs
without sacrificing performance



The Hidden Cost Driver in AI

As AI projects scale, storage demands typically outpace compute resources, creating unexpected budget challenges:

- Training datasets grow exponentially with each iteration
- Model checkpoints accumulate rapidly during experimentation
- Logs and debugging artifacts consume significant space
- Finished models require versioning and redundancy



Without strategic planning, these storage requirements can silently drain your budget while your attention is focused elsewhere.

Common Storage Cost Drivers



Excessive Hot Storage

Keeping all data in high-performance tiers regardless of access patterns



Dataset Redundancy

Multiple teams maintaining identical copies of large training datasets



Inefficient Versioning

Storing complete copies of models rather than incremental changes



Transfer Fees

Unexpected costs from moving data between regions or availability zones

These inefficiencies compound silently, often becoming apparent only after budget overruns.



Implement a Tiered Storage Approach

1

Hot Storage

Purpose: Active datasets and current model training

Characteristics: High performance, low latency, highest cost

Examples: NVMe storage, premium SSD, high IOPS block storage

2

Warm Storage

Purpose: Recent projects, validated datasets, successful models

Characteristics: Balanced performance/cost, moderate retrieval time

Examples: Standard SSDs, object storage with frequent access tiers

3

Cold Storage

Purpose: Archival, compliance, infrequently accessed backups

Characteristics: Slow retrieval, minimal cost, possible retrieval fees

Examples: Glacier, Archive storage, tape systems

Compression & Deduplication Strategies

Space-Saving Techniques

- **Lossless compression** for checkpoints and logs (10-30% savings)
- **Format-specific compression** for datasets (JPEG → WebP, etc.)
- **Delta storage** for model versions (store changes only)
- **Content-based deduplication** across research teams



Modern compression algorithms can reduce storage needs by 20-60% with minimal performance impact when implemented correctly.

Case Study: A research team reduced their storage costs by 47% by implementing automatic compression for all model checkpoints and logs while maintaining the same training performance.

Cloud-Native Cost Optimization Tools

1 — AWS

- S3 Intelligent-Tiering
- S3 Lifecycle Policies
- Glacier Deep Archive
- Cost Explorer for monitoring

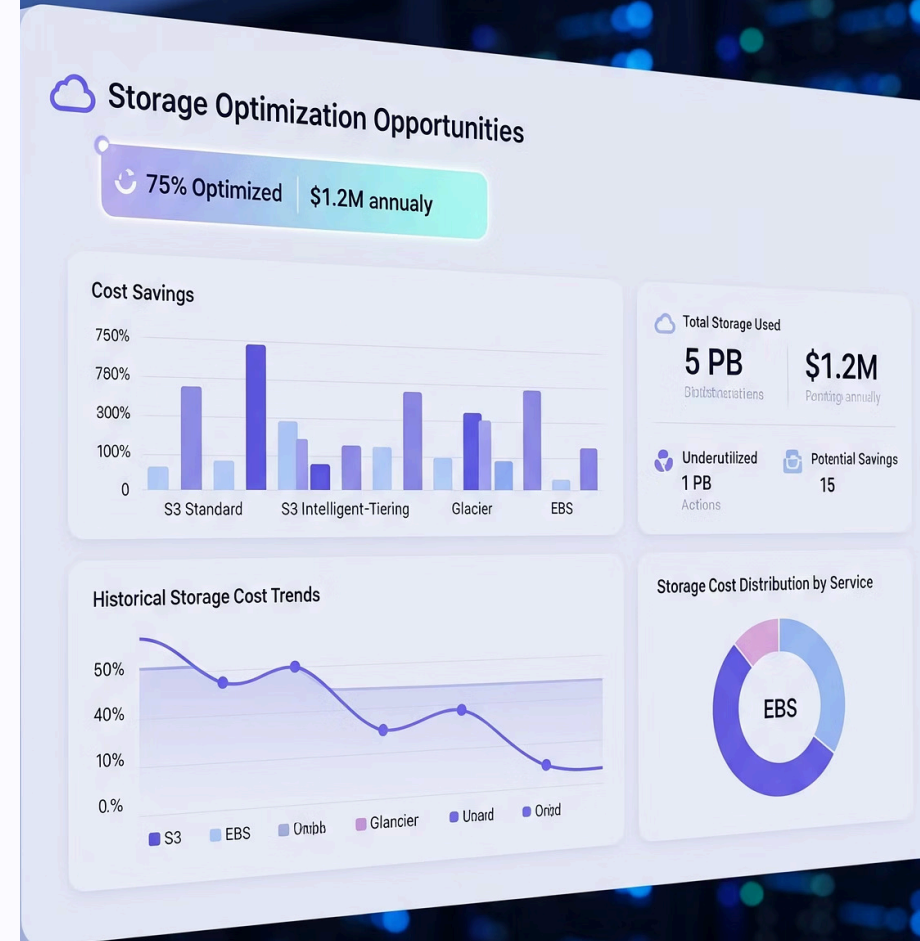
2 — Google Cloud

- Storage Transfer Service
- Object Lifecycle Management
- Autoclass for automatic tiering
- Cost Management dashboard

3 — Azure

- Blob Storage access tiers
- Immutable storage policies
- Archive Storage
- Cost Analysis tools

These platform-native tools offer the most cost-effective approach with minimal management overhead.



Implementation Best Practices

1 Establish Regular Storage Audits

Schedule quarterly reviews to identify and archive unused datasets and artifacts. Track usage patterns to inform tiering decisions.

2 Automate Lifecycle Management

Create rules that automatically move data between tiers based on age and access patterns. No human intervention required.

3 Implement Shared Feature Stores

Centralize commonly used features and embeddings to eliminate redundant storage across teams and projects.

4 Factor All Storage Costs into Planning

Include not just storage capacity but also operation costs like egress fees, retrieval charges, and cross-region transfers in your budgets.

Key Takeaways

Storage costs compound silently

What starts as a small inefficiency can grow exponentially as your AI projects scale

Tiering + compression = major savings

Implementing these two strategies alone can reduce storage costs by 30-60%

Automation prevents technical debt

Manual cleanup is rarely prioritized; build automation from the start

Strategic storage enables sustainable AI

Well-designed storage infrastructure scales with your needs without budget surprises



Questions?